GPT-6.1 Sol

10MAA End of Year Revision Examination 2026

Marking Schedule

General comments

Accept equivalent methods and exact answers unless a particular form or accuracy is requested. Require working for method marks. Correct algebraic expressions may themselves show the method. Allow follow-through where a subsequent method is valid. Do not penalise the same arithmetic error twice. Penalise incorrect rounding only once across the paper. Retain unrounded values in calculations.

Algebraic Expressions & Equations • 31 marks

Question	Working and answer	Marks	Comments
1.(a)	$(x-4)^2 = (x-4)(x-4)$ $= x^2 - 8x + 16$	2	1 for two correct cross-products or equivalent expansion. 1 for final expression.
(b)	$\dfrac{x^2-16}{x-4} = \dfrac{(x-4)(x+4)}{x-4} = x+4$	2	1 for difference-of-squares factorisation. 1 for cancellation and $x+4$.
(c)	$x^2 - 8x + 16 = x^2 + 16$ $-8x = 0$, so $\boldsymbol{x = 0}$.	2	1 for equating the correct expansion to the student's expression. 1 for 0.
2.(a)	$(16m^{12})^{\frac{1}{4}} = 2m^3$	2	1 for fourth root of 16. 1 for m^3.
(b)	$3^{2a} \times 3^{3b} = 3^{22}$ $2a + 3b = 22$, $a = b + 1$ $2(b+1) + 3b = 22$ $\boldsymbol{b = 4}$, $\boldsymbol{a = 5}$.	3	1 for a common base. 1 for the equation in one variable. 1 for both values.
3.(a)	First differences: $7, 11, 15$. Second differences: $4, 4$, which are constant.	2	1 for all first differences. 1 for second differences and conclusion.
(b)	$T = an^2 + bn + c$ $2a = 4$, so $a = 2$. $2 + b + c = 7$, $8 + 2b + c = 14$ $b = 1$, $c = 4$ $\boldsymbol{T = 2n^2 + n + 4}$.	3	1 for quadratic coefficient 2. 1 for finding the remaining coefficients. 1 for the complete formula. Accept other valid methods.
(c)	$T(10) = 2(100) + 10 + 4 =$ **214 ms**.	1	Allow follow-through from (b).
4.(a)	$p + r = 20$ $3p + 5r = 78$	2	1 for each equation.
(b)	$3p + 3r = 60$ $2r = 18$, so $r = 9$ and $p = 11$. **11 Python stickers and 9 Rust stickers**.	2	1 for valid elimination or substitution. 1 for both counts, correctly identified.

Algebraic Expressions & Equations continued			
Question	**Working and answer**	**Marks**	**Comments**
5.	$2(3x-5) > 5x + k$ gives $x > k + 10$. True at $x = 4$: $4 > k + 10$, so $k \leftarrow 6$. False at $x = 1$: $1 \leq k + 10$, so $k \geq -9$. Thus $-9 \leq k \leftarrow 6$. $\boldsymbol{k = -9, -8, -7}$.	4	1 for correct simplification or substitution at both test values. 1 for upper bound $k \leftarrow 6$. 1 for lower bound $k \geq -9$. 1 for the complete integer list.
6.(a)	The enlarged dimensions are $(n + 2)$ and $(n + 9)$. $\boldsymbol{(n+2)(n+9) = 198}$.	2	1 for both enlarged dimensions. 1 for the area equation.
(b)	$n^2 + 11n - 180 = 0$ $(n + 20)(n - 9) = 0$ $n = -20$ or $n = 9$. Reject -20 because a cell count cannot be negative. Original dimensions: **9 cells by 16 cells**.	4	1 for the correct quadratic. 1 for solving it. 1 for selecting 9 with a contextual reason. 1 for both original dimensions.

Statistics & Probability • 27 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	$36 - 18 =$ **18 seconds**.	1	Correct difference.
(b)	Linux: $22 - 16 = 6$ s. macOS: $30 - 20 = 10$ s. **Linux has the smaller interquartile range**.	2	1 for both interquartile ranges. 1 for comparison identifying Linux.
(c)	macOS: **approximately 75%**. Linux: **100%**.	2	1 for each. Accept a quartile-based estimate for macOS.
(d)	**No**. The slowest Linux run is 28 s, while the fastest Windows run is 20 s.	1	A valid comparison or counterexample is required. Merely stating that the median is lower does not establish the claim.
2.(a)	$2 + 5 + 7 + 9 + k + 15 = 6(8)$ $38 + k = 48$, so $\boldsymbol{k = 10}$.	2	1 for using total $= 6 \times \text{mean}$. 1 for 10.
(b)	Median before: $\dfrac{7 + 9}{2} = 8$. Median after: **8**. Changing the largest value does not change the third and fourth values.	2	1 for both medians. 1 for explaining that the middle two values are unchanged.

Statistics & Probability continued			
Question	**Working and answer**	**Marks**	**Comments**
3.(a)	First card: Rust $\frac{3}{9}$; not Rust $\frac{6}{9}$. After Rust: Rust $\frac{2}{8}$; not Rust $\frac{6}{8}$. After not Rust: Rust $\frac{3}{8}$; not Rust $\frac{5}{8}$.	3	1 for each complete pair of branches on a correctly structured two-stage tree. Accept simplified fractions.
(b)	$P(\text{same}) = \frac{4}{9}\times\frac{3}{8} + \frac{3}{9}\times\frac{2}{8} + \frac{2}{9}\times\frac{1}{8}$ $= \frac{12+6+2}{72} = \frac{20}{72} = \mathbf{\frac{5}{18}}$.	3	1 for correct Python and Rust paths. 1 for the C path and addition of all three. 1 for $\frac{5}{18}$. Accept a combinations method.
(c)	$P(\text{both Rust}) = \frac{3}{9}\times\frac{2}{8} = \frac{1}{12}$ $P(\text{both Rust} \mid \text{same}) = \frac{\frac{1}{12}}{\frac{5}{18}} = \mathbf{\frac{3}{10}}$.	2	1 for using the same-language event as the condition. 1 for $\frac{3}{10}$.
(d)	$180 \times \frac{5}{18} =$ **50 pairs**.	1	Allow follow-through from (b).
4.(a)	**Diagram B**. Its bar areas are $20(0.6) = 12$, $20(0.9) = 18$, $40(0.6) = 24$, $60(0.3) = 18$, so area represents frequency. In A, unequal class widths make the areas inconsistent with the frequencies.	2	1 for Diagram B. 1 for a correct area–frequency justification. Merely naming frequency density without explaining areas earns only the first mark.
(b)	$20(0.6) =$ **12 projects**. Assume build times are evenly spread within $40 < t \le 80$.	2	1 for 12. 1 for the uniform-within-class assumption.
(c)	All 18 projects in $80 < t \le 140$ exceed 50 s. Between 0 and 24 of the projects in $40 < t \le 80$ could exceed 50 s. **Least 18; greatest 42**.	2	1 for each bound. No within-class assumption is allowed here.
(d)	Let $u = \frac{t}{2}$. New class: $\mathbf{20 < u \le 40}$ **seconds**. Frequency stays 24, while width becomes 20. New density: $\mathbf{\frac{24}{20} = 1.2}$.	2	1 for the halved interval. 1 for density 1.2, not 0.3. Accept the original variable name if the meaning is clear.

Trigonometry • 13 marks

Question	Working and answer	Marks	Comments
1.(a)	$\dfrac{\sin C}{14} = \dfrac{\sin 35°}{10}$ $\sin C = 1.4 \sin 35° = 0.803007...$ $C = 53.4182...°$ or $180° - 53.4182...°$ **53.4° or 126.6°**.	3	1 for correct sine-rule equation. 1 for the acute value. 1 for the obtuse value.
(b)	$A = 180° - 35° - 126.5818...°$ $= 18.4182...°$ $\dfrac{BC}{\sin A} = \dfrac{10}{\sin 35°}$ $BC = 5.50843... =$ **5.51 cm**.	2	1 for the correct remaining angle and sine-rule setup. 1 for BC. Allow a valid alternative method.
2.(a)	$\boldsymbol{h = x \tan 55°}$.	1	Correct expression.
(b)	$\boldsymbol{x \tan 55° = (x + 20) \tan 35°}$.	1	An equation in x only is required.
(c)	$x(\tan 55° - \tan 35°) = 20 \tan 35°$ $x = 19.2380...$ $h = x \tan 55° = 27.4748...$ **27.5 m**.	3	1 for correctly isolating x. 1 for calculating h from x. 1 for final height. Allow follow-through.
3.	Bearing of A from B is 220°. $\angle ABC = 220° - 100° = 120°$. $AC^2 = 8^2 + 6^2 - 2(8)(6)\cos 120°$ $AC = \sqrt{148} =$ **12.2 km**.	3	1 for included angle 120°. 1 for cosine rule. 1 for distance.

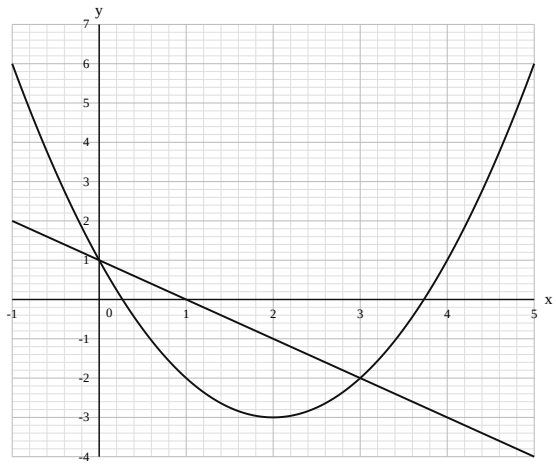
Measurement • 16 marks

Question	Working and answer	Marks	Comments
1.(a)	Internal dimensions: $12 - 0.8 = 11.2$ cm, $8 - 0.8 = 7.2$ cm, $5 - 0.4 = 4.6$ cm. $V = 11.2(7.2)(4.6) = 370.944\text{cm}^3$ **371 cm³**.	3	1 for reducing both horizontal dimensions by twice the wall thickness. 1 for internal height 4.6. 1 for capacity.
(b)	$V_{\text{external}} = 12(8)(5) = 480\text{cm}^3$ $V_{\text{material}} = 480 - 370.944 = 109.056\text{cm}^3$ **109 cm³**.	2	1 for external volume minus internal capacity. 1 for material volume.
(c)	$\text{Mass} = 109.056(1.24) = 135.22944$ **135 g**.	1	Allow follow-through from (b).

Measurement continued			
Question	**Working and answer**	**Marks**	**Comments**
2.(a)	Volume scale factor $= \dfrac{729}{216} = \dfrac{27}{8}$ Length scale factor $= \sqrt[3]{\dfrac{27}{8}} = \dfrac{3}{2}$ Height $= 8 \times \dfrac{3}{2} =$ **12 cm**.	2	1 for the cube root giving length scale factor 1.5. 1 for 12 cm.
(b)	Area scale factor $= \left(\dfrac{3}{2}\right)^2 = \dfrac{9}{4}$ Area $= 96 \times \dfrac{9}{4} =$ **216 cm²**.	2	1 for squaring the length scale factor. 1 for area.
3.(a)	$\mathbf{2.95 \le r < 3.05}$ **cm**. $\mathbf{11.5 \le h < 12.5}$ **cm**.	2	1 for each complete pair of bounds.
(b)	$V_{\text{minimum}} = \pi(2.95)^2(11.5)$ $= 314.406...\text{cm}^3 < 350\text{cm}^3$. **The claim is not guaranteed**.	2	1 for a valid lower-bound capacity calculation. 1 for comparison with 350 and conclusion. An attainable counterexample is also valid.
(c)	$V_{\text{water}} = \pi(3)^2(12) - \dfrac{1}{3}\pi(3)^2(12)$ $= 108\pi - 36\pi = 72\pi\text{cm}^3$ **226 cm³**.	2	1 for cylinder volume minus cone volume. 1 for 72π or 226 cm³.

Coordinate Geometry • 12 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	$m_{AB} = \dfrac{5-1}{4-(-4)} = \dfrac{1}{2}$ $m_{BC} = \dfrac{1-5}{6-4} = -2$ $m_{AB}m_{BC} = -1$, so $AB \perp BC$.	3	1 for each gradient. 1 for product -1 and perpendicular conclusion.
(b)	AC is horizontal, with length $6-(-4) = 10$. The perpendicular height from B is $5-1 = 4$. $A = \dfrac{1}{2}(10)(4) =$ **20 square units**.	2	1 for a valid base and height, or both perpendicular side lengths. 1 for area.
(c)	Reflection in $y = 1$ gives $\mathbf{(4, -3)}$.	1	Correct coordinates.
2.(a)	$2x + 1 = -\dfrac{1}{2}x + 6$ $\dfrac{5}{2}x = 5$, so $x = 2$, $y = 5$. **P** $\mathbf{(2, 5)}$.	2	1 for solving the simultaneous equations. 1 for both coordinates.
(b)	At A: $-3 = 2x + 1$, so $x = -2$. At B: $-3 = -\dfrac{1}{2}x + 6$, so $x = 18$. $AB = 18 - (-2) =$ **20 units**.	2	1 for both x-coordinates. 1 for distance.
(c)	$AK = 2-(-2) = 4$, $KB = 18-2 = 16$. The triangles have equal height PK. Area ratio $= 4 : 16 =$ $\mathbf{1 : 4}$.	2	1 for correct bases and common-height reasoning, or correct individual areas 16 and 64. 1 for ratio 1 : 4.

Graphs of Quadratic Functions • 23 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	$y = a(x-2)^2 + 9$ At $(0, 5)$: $5 = 4a + 9$, so $a = -1$. $\boldsymbol{y = -(x-2)^2 + 9}$.	3	1 for the correct vertex form. 1 for substituting $(0, 5)$ and finding a. 1 for the equation.
(b)	$0 = -(x-2)^2 + 9$ $(x-2)^2 = 9$, so $x = 2 \pm 3$. $\boldsymbol{x = -1}$ **m and** $\boldsymbol{x = 5}$ **m**. Both are valid positions: 1 m west and 5 m east of the post.	3	1 for setting $y = 0$ and rearranging. 1 for both roots. 1 for retaining and interpreting both positions. Accept $(-1, 0)$ and $(5, 0)$.
(c)	$-(x-2)^2 + 9 \geq 8$ $(x-2)^2 \leq 1$ $\boldsymbol{1 \leq x \leq 3}$.	3	1 for the height inequality. 1 for boundary values 1 and 3. 1 for the correct closed interval between them.
2.(a)	$2(n^2 + 3n) + 12$ $= 2\left(\left(n + \frac{3}{2}\right)^2 - \frac{9}{4}\right) + 12$ $\boldsymbol{2\left(n + \frac{3}{2}\right)^2 + \frac{15}{2}}$.	2	1 for the correct square term. 1 for constant $\frac{15}{2}$. Accept decimal equivalents.
(b)	$2n^2 + 6n + 12 \leq 200$ Boundary: $n^2 + 3n - 94 = 0$. $n = \frac{-3 \pm \sqrt{385}}{2}$ $n = -11.3107...$ or $n = 8.3107....$ Since n is a positive whole number, **8 items**. Check: $M(8) = 188 \leq 200$; $M(9) = 228 > 200$.	4	1 for the capacity inequality or boundary equation. 1 for solving the quadratic. 1 for applying the positive whole-number domain. 1 for 8, justified by the interval or the neighbouring memory values.
(c)	A **translation** $\frac{3}{2}$ units left and $\frac{15}{2}$ units up.	2	1 for each correctly directed component.

Graphs of Quadratic Functions continued			
Question	**Working and answer**	**Marks**	**Comments**
3.(a)		2	1 for at least two correctly plotted points on $y = 1 - x$. 1 for a straight line through them across the grid.
(b)	$\mathbf{(0, 1)}$ **and** $\mathbf{(3, -2)}$.	2	1 for each coordinate pair. Allow sensible graph-reading tolerance.
(c)	$x^2 - 4x + 1 = 1 - x$ $x^2 - 3x = 0$ $x(x - 3) = 0$, so $\boldsymbol{x = 0}$ **or** $\boldsymbol{x = 3}$.	2	1 for equating the functions and rearranging. 1 for both roots.

Geometry • 38 marks			
Question	**Working and answer**	**Marks**	**Comments**
1.(a)	$OA = OB$ because both are radii. $\angle OAB = \dfrac{180° - 110°}{2} = \mathbf{35°}$.	2	1 for 35°. 1 for equal base angles in an isosceles triangle and the triangle angle sum.
(b)	$\angle OAT = \angle OBT = 90°$. $\angle ATB = 360° - 110° - 90° - 90°$ $\boldsymbol{\angle ATB = 70°}$.	3	1 for the radius–tangent right angles. 1 for using the angle sum of quadrilateral OATB. 1 for 70°.
(c)	C lies on the minor arc AB, so angle ACB subtends the major arc AB. The corresponding reflex central angle is $360° - 110° = 250°$. $\angle ACB = \dfrac{1}{2}(250°) = \mathbf{125°}$.	2	1 for 125°. 1 for the angle at the circumference being half the central angle on the same arc, correctly using the major arc.
2.(a)	$\overrightarrow{AB} = \boldsymbol{q} - \boldsymbol{p}$ $\overrightarrow{OC} = \boldsymbol{p} + \dfrac{2}{3}(\boldsymbol{q} - \boldsymbol{p})$ $\boldsymbol{\overrightarrow{OC} = \frac{1}{3}p + \frac{2}{3}q}$.	2	1 for the correct fraction of AB. 1 for OC.
(b)	$\overrightarrow{AC} = \dfrac{2}{3}(\boldsymbol{q} - \boldsymbol{p})$ $\overrightarrow{AD} = (2\boldsymbol{q} - \boldsymbol{p}) - \boldsymbol{p}$ $= 2(\boldsymbol{q} - \boldsymbol{p}) = 3\overrightarrow{AC}$. The vectors share starting point A and are parallel, so A, C and D are collinear.	2	1 for a correct comparison of AC and AD. 1 for the common-starting-point collinearity conclusion.
(c)	$\overrightarrow{CD} = \overrightarrow{AD} - \overrightarrow{AC} = 2\overrightarrow{AC}$ $\boldsymbol{AC : CD = 1 : 2}$.	2	1 for CD being twice AC. 1 for ratio 1 : 2.

Geometry continued • additional vector questions			
Question	**Working and answer**	**Marks**	**Comments**
3.(a)	$\overrightarrow{OP} = \frac{2}{3}\boldsymbol{p}$ $\overrightarrow{OQ} = \frac{2}{3}\boldsymbol{q}$	2	1 for each vector.
(b)	$\overrightarrow{OR} = \frac{1}{2}\left(\overrightarrow{OA} + \overrightarrow{OQ}\right)$ $= \frac{1}{2}\boldsymbol{p} + \frac{1}{3}\boldsymbol{q}$ $\overrightarrow{OS} = \frac{1}{2}\left(\overrightarrow{OB} + \overrightarrow{OP}\right)$ $= \frac{1}{3}\boldsymbol{p} + \frac{1}{2}\boldsymbol{q}$	2	1 for each midpoint position vector. Accept equivalent unsimplified expressions.
(c)	$\overrightarrow{RS} = \overrightarrow{OS} - \overrightarrow{OR}$ $= \frac{1}{6}(\boldsymbol{q} - \boldsymbol{p})$ $\overrightarrow{AB} = \boldsymbol{q} - \boldsymbol{p}$ Thus $\overrightarrow{RS} = \frac{1}{6}\overrightarrow{AB}$. $\boldsymbol{RS \parallel AB}$ **and** $\boldsymbol{RS : AB = 1 : 6}$.	2	1 for the correct vector RS. 1 for comparison with AB, the parallel conclusion and ratio $1 : 6$.
4.(a)	$\overrightarrow{PQ} = \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$ $\overrightarrow{SR} = \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$	2	1 for each vector. Direction SR, not RS, is required.
(b)	$\overrightarrow{PQ} = \overrightarrow{SR}$, so PQ and SR have equal length and are parallel in the same direction. One pair of opposite sides of PQRS is equal and parallel. **PQRS is a parallelogram**.	2	1 for the directed-vector comparison. 1 for the correct parallelogram criterion and conclusion.
(c)	From A: $\overrightarrow{AP} = \frac{1}{2}\boldsymbol{a}$, $\overrightarrow{AR} = \frac{1}{2}\boldsymbol{a} + \boldsymbol{b}$; midpoint of PR has vector $\frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$. $\overrightarrow{AQ} = \boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$, $\overrightarrow{AS} = \frac{1}{2}\boldsymbol{b}$; midpoint of QS has vector $\frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b}$. Since $\overrightarrow{AC} = \boldsymbol{a} + \boldsymbol{b}$, both are the midpoint of AC.	2	1 for the midpoint of PR. 1 for the midpoint of QS and comparison with the midpoint of AC. A statement that diagonals bisect each other without the requested position-vector working is insufficient.

Geometry continued • additional vector questions			
Question	**Working and answer**	**Marks**	**Comments**
5.(a)	$\overrightarrow{AP} = \boldsymbol{a} + \frac{2}{5}(\boldsymbol{b} - \boldsymbol{a})$ $= \frac{3}{5}\boldsymbol{a} + \frac{2}{5}\boldsymbol{b}$ $\overrightarrow{BQ} = -\boldsymbol{a} + \frac{4}{5}\boldsymbol{b}$	2	1 for each vector.
(b)	$\overrightarrow{AT} = k\left(\frac{3}{5}\boldsymbol{a} + \frac{2}{5}\boldsymbol{b}\right)$ $\overrightarrow{AT} = \boldsymbol{a} + m\left(-\boldsymbol{a} + \frac{4}{5}\boldsymbol{b}\right)$ Comparing coefficients: $\boldsymbol{\frac{3}{5}k = 1 - m}$ and $\boldsymbol{\frac{2}{5}k = \frac{4}{5}m}$.	2	1 for each correct coefficient equation, supported by the two vector expressions. Allow follow-through from (a).
(c)	$m = \frac{k}{2}$ $\frac{3}{5}k = 1 - \frac{k}{2}$, so $\frac{11}{10}k = 1$. $k = \frac{10}{11}$, so $AT = \frac{10}{11}AP$ and $TP = \frac{1}{11}AP$. $\boldsymbol{AT : TP = 10 : 1}$.	2	1 for $k = \frac{10}{11}$. 1 for ratio 10 : 1, not 10 : 11.
6.(a)	$\overrightarrow{OB} = 2\boldsymbol{a} + \boldsymbol{b}$ $\overrightarrow{AC} = -6\boldsymbol{a} + \boldsymbol{b}$	2	1 for each vector.
(b)	$\overrightarrow{OX} = s(2\boldsymbol{a} + \boldsymbol{b})$ $\overrightarrow{OX} = 6\boldsymbol{a} + t(-6\boldsymbol{a} + \boldsymbol{b})$ Coefficients of $\boldsymbol{a}$: $2s = 6 - 6t$. Coefficients of $\boldsymbol{b}$: $s = t$. $8s = 6$, hence $\boldsymbol{s = t = \frac{3}{4}}$.	3	1 for each coefficient equation derived from the two vector routes. 1 for solving and stating both parameters.
(c)	$\overrightarrow{OX} = \frac{3}{2}\boldsymbol{a} + \frac{3}{4}\boldsymbol{b}$. As $MN \parallel OA$, $\overrightarrow{OM} = \frac{3}{4}\boldsymbol{b}$. Therefore $\boldsymbol{\overrightarrow{MX} = \frac{3}{2}a}$. On AB, write $\overrightarrow{ON} = 6\boldsymbol{a} + u(-4\boldsymbol{a} + \boldsymbol{b})$. Its $\boldsymbol{b}$ coefficient is $\frac{3}{4}$, so $u = \frac{3}{4}$ and $\overrightarrow{ON} = 3\boldsymbol{a} + \frac{3}{4}\boldsymbol{b}$. Therefore $\boldsymbol{\overrightarrow{XN} = \frac{3}{2}a}$. Since $\overrightarrow{MX} = \overrightarrow{XN}$, **X is the midpoint of MN**.	2	1 for MX. 1 for XN with a valid midpoint conclusion. Accept an equivalent method using segment ratios.

Total: 160 marks